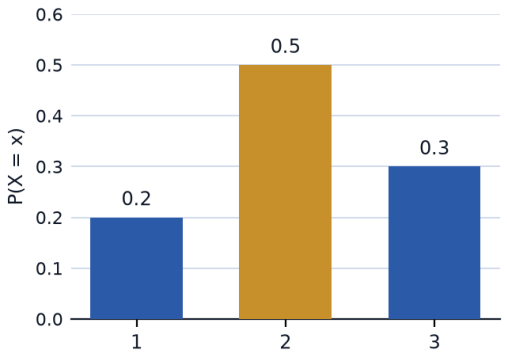
Discrete Random Variables

ClipSAT Printable Practice Suite (clipsat.org) • Topic 17.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

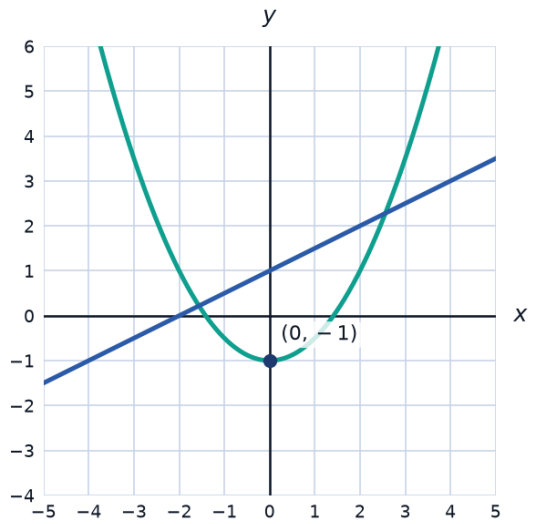
1 [Foundational] • Target: Expected value (mean) of a discrete random variable

A random variable X has distribution $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. Find $E(X)$.



2 [Foundational] • Target: Probability distributions

A discrete random variable X has the probability distribution: $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, $P(X = 3) = k$, $P(X = 4) = 0.1$. Find k, using the fact that probabilities must sum to 1.



3 [Foundational] • Target: Probability distributions

A discrete random variable Y has the probability distribution: $P(Y = 0) = 0.1$, $P(Y = 1) = k$, $P(Y = 2) = 2k$, $P(Y = 3) = 0.3$. Find k.

4 [Foundational] • Target: Probability distributions

A discrete random variable Z has the probability distribution: $P(Z = 1) = 0.25$, $P(Z = 2) = 0.25$, $P(Z = 3) = k$, $P(Z = 4) = 0.15$, $P(Z = 5) = 0.15$. Find k.

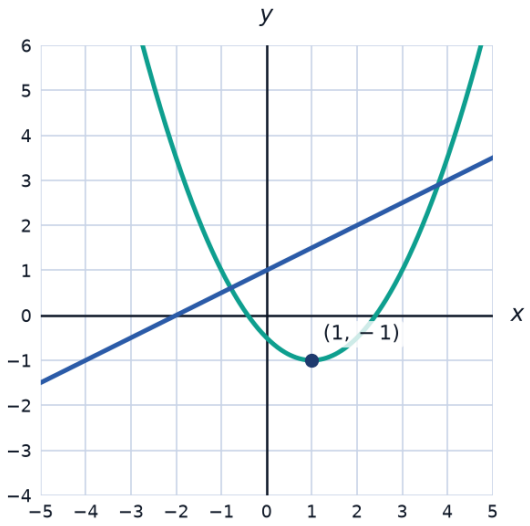
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5 [Foundational] • Target: Expected value (mean) of a discrete random variable

A random variable Y has distribution $P(Y=0)=0.1$, $P(Y=1)=0.4$, $P(Y=2)=0.5$. Find $E(Y)$.

6 [Foundational] • Target: Expected value (mean) of a discrete random variable

A random variable Z has distribution $P(Z=1)=0.3$, $P(Z=2)=0.3$, $P(Z=3)=0.4$. Find $E(Z)$.



7 [Foundational] • Target: Variance of a discrete random variable

Using the distribution from the previous question ($E(X)=2.1$), find $\text{Var}(X)$, using $\text{Var}(X)=E(X^2)-[E(X)]^2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Variance of a discrete random variable

Using the distribution $P(Y=0)=0.1$, $P(Y=1)=0.4$, $P(Y=2)=0.5$ ($E(Y)=1.4$), find $\text{Var}(Y)$, using $\text{Var}(Y)=E(Y^2)-[E(Y)]^2$.

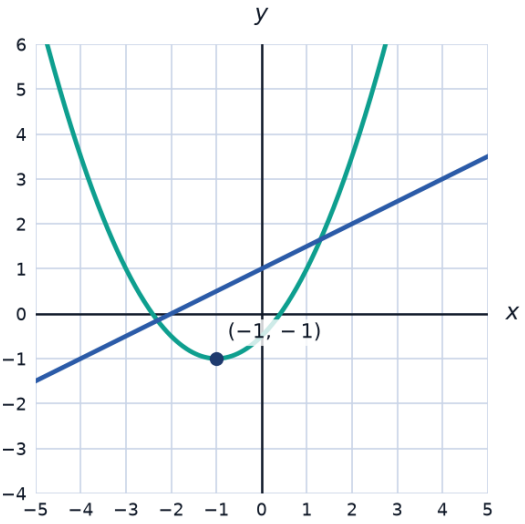
9 [Intermediate] • Target: Variance of a discrete random variable

Using the distribution $P(Z=1)=0.3$, $P(Z=2)=0.3$, $P(Z=3)=0.4$ ($E(Z)=2.1$), find $\text{Var}(Z)$, using $\text{Var}(Z)=E(Z^2)-[E(Z)]^2$.

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10 [Intermediate] • Target: Linear transformations of a random variable

A random variable X has $E(X) = 5$ and $\text{Var}(X) = 3$. Find $E(2X + 4)$ and $\text{Var}(2X + 4)$.



11 [Intermediate] • Target: Linear transformations of a random variable

A random variable X has $E(X) = 8$ and $\text{Var}(X) = 5$. Find $E(3X - 2)$ and $\text{Var}(3X - 2)$.

12 [Intermediate] • Target: Linear transformations of a random variable

A random variable Y has $E(Y) = 10$ and $\text{Var}(Y) = 4$. Find $E(5 - 2Y)$ and $\text{Var}(5 - 2Y)$.

13 [Intermediate] • Target: The discrete uniform distribution

A fair six-sided die is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, given that for a discrete uniform distribution on $1, \ldots, n$: $E(X) = \frac{n+1}{2}$ and $\text{Var}(X) = \frac{n^2-1}{12}$.

14 [Intermediate] • Target: The discrete uniform distribution

A fair eight-sided die (labeled 1 to 8) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

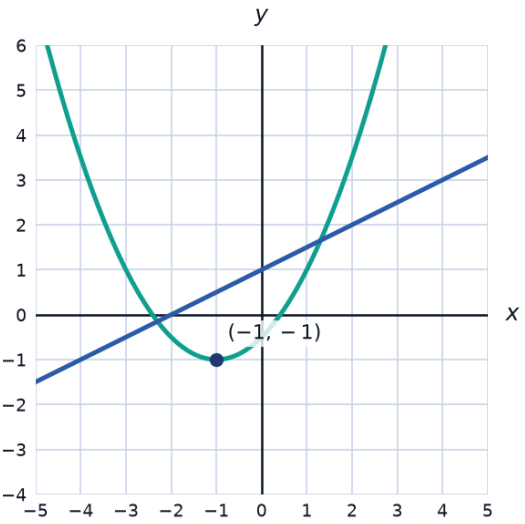
15 [Challenging] • Target: The discrete uniform distribution • Cross-Lesson: 13.01 Newton's Laws

A fair ten-sided die (labeled 1 to 10) is rolled, and X is the number shown. Find $E(X)$ and $\text{Var}(X)$, using the same formulas.

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16 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 14.01 Energy, Work & Power

A game involves rolling two fair dice and finding the sum X. Find $P(X = 4)$ and $P(X = 7)$.



17 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 15.01 Representation of Data

Using the same game of rolling two fair dice and finding the sum X, find $P(X = 9)$ and $P(X = 12)$.

18 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 16.01 Permutations & Combinations

Using the same game of rolling two fair dice and finding the sum X, find $P(X = 5)$ and $P(X = 6)$.

19 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 18.01 Binomial Distribution

A random variable X has $P(X = 1) = 0.15$, $P(X = 2) = 0.35$, $P(X = 3) = 0.25$, $P(X = 4) = 0.25$. Find $P(X \leq 2)$ and $P(X > 2)$.

20 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 19.01 Normal Distribution

A random variable X has $P(X = 1) = 0.1$, $P(X = 2) = 0.2$, $P(X = 3) = 0.3$, $P(X = 4) = 0.4$. Find $P(X \leq 3)$ and $P(X \geq 2)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Probability Distribution Tables & Total Probability

The random variable X has probability distribution $P(X = x) = c(x^2 + 1)$ for $x = 0, 1, 2, 3$. Find the constant c.

a. $\frac{1}{18}$

b. $\frac{1}{14}$

c. $\frac{1}{15}$

d. $\frac{1}{20}$

22 [Foundational] • Target: Probability Distribution Tables & Total Probability

A fair die is rolled. Let X be the number of rolls until a 6 appears. What kind of distribution does X follow?

a. Geometric distribution

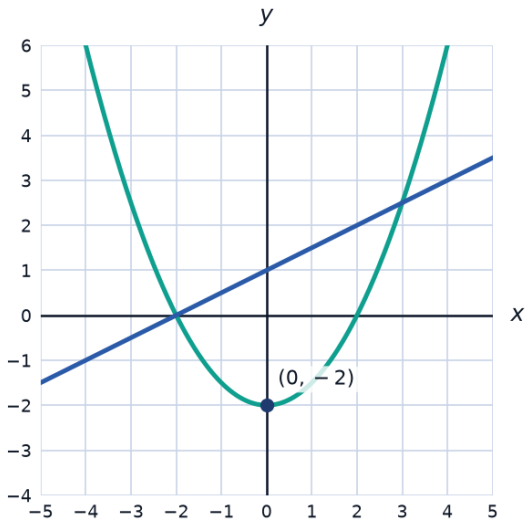
b. Binomial distribution

c. Uniform distribution

d. Poisson distribution

23 [Foundational] • Target: Probability Distribution Tables & Total Probability

Given the probability distribution of Y: $P(Y = -1) = 0.3, P(Y = 0) = 0.4, P(Y = 2) = 0.3$. Find $P(Y \geq 0)$.



a. 0.7

b. 0.4

c. 0.3

d. 1.0

24 [Foundational] • Target: Expected Value $E(X)$ & Moments

Find the expectation $E(X)$ for the distribution: $P(X = 1) = 0.1, P(X = 2) = 0.3, P(X = 3) = 0.4, P(X = 4) = 0.2$.

a. 2.7

b. 2.5

c. 3.0

d. 2.8

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25 [Foundational] • Target: Expected Value $E(X)$ & Moments

If $E(X) = 5$, what is $E(3X - 4)$?

a. 11

b. 15

c. 19

d. 7

26 [Foundational] • Target: Expected Value $E(X)$ & Moments

A game charges 5 dollars to play. A fair coin is tossed: Heads pays 8 dollars, Tails pays nothing. What is the expected net gain for the player?

a. -1.00

b. +1.00

c. -2.00

d. +4.00

27 [Foundational] • Target: Variance $\text{Var}(X)$ & Standard Deviation

Given $E(X) = 3$ and $E(X^2) = 13$, find $\text{Var}(X)$.

a. 4

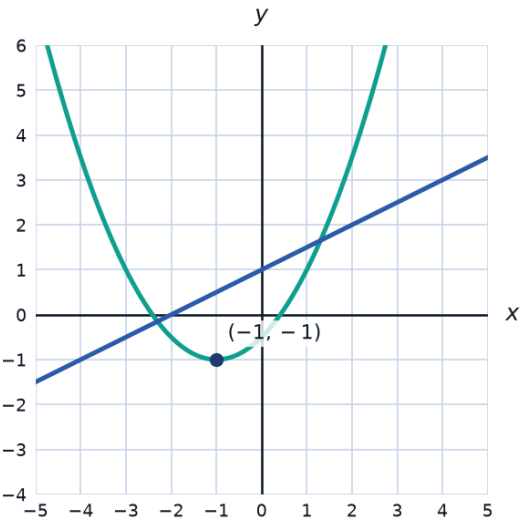
b. 10

c. 2

d. 16

28 [Foundational] • Target: Variance $\text{Var}(X)$ & Standard Deviation

If $\text{Var}(X) = 6$, find the variance of $Y = 2X - 5$.



a. 24

b. 12

c. 7

d. 36

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Variance $\text{Var}(X)$ & Standard Deviation**

Calculate the standard deviation of X given $P(X = 0) = 0.5$ and $P(X = 2) = 0.5$.

a. 1

b. 2

c. 0.5

d. $\sqrt{2}$

30 **[Intermediate]** • **Target: Multi-Stage Discrete Problem Solving**

Two fair 4-sided dice (numbered 1 to 4) are rolled. Let S be the sum of the two numbers. Find $P(S > 5)$.

a. $\frac{6}{16} = \frac{3}{8}$

b. $\frac{1}{2}$

c. $\frac{5}{16}$

d. $\frac{1}{4}$

31 **[Intermediate]** • **Target: Multi-Stage Discrete Problem Solving**

For the sum S of the two 4-sided dice in the previous question, what is $E(S)$?

a. 5

b. 4

c. 6

d. 4.5

32 **[Intermediate]** • **Target: Multi-Stage Discrete Problem Solving**

A box has 3 red and 2 blue balls. Two balls are drawn without replacement. Let X be the number of blue balls drawn. Find $\text{Var}(X)$.

a. 0.36

b. 0.40

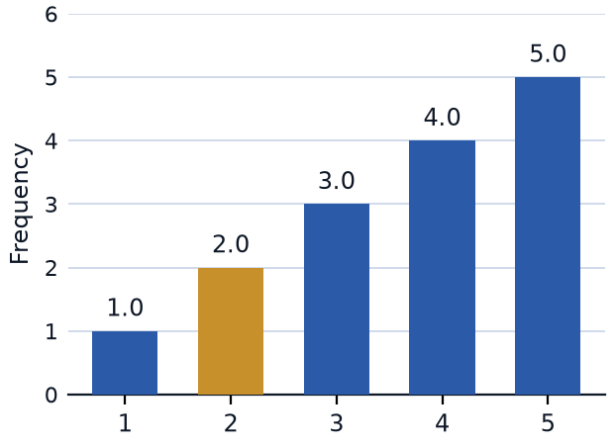
c. 0.48

d. 0.24

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33 [Intermediate] • Target: Probability distributions

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.



a. $\frac{1}{15}$

b. $\frac{1}{5}$

c. $\frac{1}{3}$

d. 1

34 [Intermediate] • Target: Expected value (mean) of a discrete random variable

A discrete random variable X has distribution $P(X = 0) = 1/3, P(X = 1) = 1/2, P(X = 2) = 1/12, P(X = 3) = 1/12$. Find $\text{Var}(X)$.

a. 3/20

b. 107/144

c. 11/34

d. 1

35 [Intermediate] • Target: Variance of a discrete random variable

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/2, P(X = 3) = 1/10$. Find $\text{Var}(X)$.

a. 3/5

b. 8/9

c. 41/100

d. 6/7

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36 [Intermediate] • Target: Linear transformations of a random variable

A discrete random variable X has distribution $P(X=1)=3/5, P(X=2)=1/10, P(X=3)=3/10$. Find $\mathrm{Var}(X)$.

a. 7/11

b. 7/10

c. 81/100

d. 11/21

37 [Intermediate] • Target: The discrete uniform distribution

A discrete random variable X has distribution $P(X=1)=4/5, P(X=2)=1/10, P(X=3)=1/10$. Find $\mathrm{Var}(X)$.

a. 1/2

b. 41/100

c. 4/5

d. 1/6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Constructing a probability distribution from a real scenario • Cross-Lesson: 16.01 Permutations & Combinations

A discrete random variable X has distribution $P(X=1)=3/10, P(X=2)=1/10, P(X=3)=3/5$. Find $\mathrm{Var}(X)$.

a. 15/28

b. 2/5

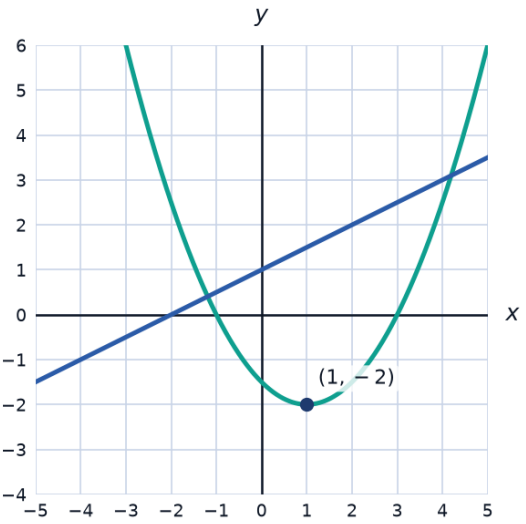
c. 25/34

d. 81/100

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39 [Challenging] • Target: Cumulative distribution function • Cross-Lesson: 18.01 Binomial Distribution

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 3/10, P(X = 3) = 3/10$. Find $\text{Var}(X)$.



a. 12/35

b. 69/100

c. 5/12

d. 13/34

40 [Challenging] • Target: A well-designed word problem: a raffle ticket game • Cross-Lesson: 19.01 Normal Distribution

A discrete random variable X has distribution $P(X = 1) = 1/5, P(X = 2) = 3/10, P(X = 3) = 1/2$. Find $\text{Var}(X)$.

a. 61/100

b. 8/17

c. 6/19

d. 1/9

41 [Challenging] • Target: Probability Distribution Tables & Total Probability • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A discrete random variable X has distribution $P(X = -3) = 7/12, P(X = -1) = 1/12, P(X = 1) = 1/12, P(X = 4) = 1/4$. Find $\text{Var}(X)$.

a. 425/48

b. 38/41

c. 11/35

d. 1

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42 [Challenging] • Target: Expected Value $E(X)$ & Moments • Cross-Lesson: 3.01 Graphs & Transformations

A discrete random variable X has distribution $P(X=0)=1/6, P(X=1)=1/12, P(X=2)=5/12, P(X=3)=1/3$. Find $\mathrm{Var}(X)$.

a. 1

b. 31/35

c. 155/144

d. 1/16

43 [Challenging] • Target: Variance $\text{Var}(X)$ & Standard Deviation • Cross-Lesson: 4.01 Coordinate Geometry

A discrete random variable X has distribution $P(X=-2)=1/5, P(X=0)=2/5, P(X=3)=1/10, P(X=4)=3/10$. Find $\mathrm{Var}(X)$.

a. 529/100

b. 40/43

c. 20/41

d. 5/13

44 [Challenging] • Target: Multi-Stage Discrete Problem Solving • Cross-Lesson: 5.01 Trigonometry

A discrete random variable X has distribution $P(X=-3)=1/10, P(X=0)=3/20, P(X=2)=1/20, P(X=5)=7/10$. Find $\mathrm{Var}(X)$.

a. 771/100

b. 2/41

c. 25/31

d. 5/7

45 [Challenging] • Target: Probability distributions • Cross-Lesson: 6.01 Differentiation & Integration

A discrete random variable X has distribution $P(X=1)=1/2, P(X=2)=3/10, P(X=3)=1/5$. Find $\mathrm{Var}(X)$.

a. 61/100

b. 7/12

c. 1/19

d. 15/16

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